CTFS
连续→离散　$x(t)\to a_k$

$$\begin{cases} x(t)=\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} & \omega_0=\frac{2\pi}{T} \\ a_k=\frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt \end{cases}$$

DTFS
离散→离散　$x[n]\to a_k$

$$\begin{cases} x[n]=\sum_{k=0}^{N-1} a_k e^{jk\omega_0 n} & \text{基波周期}N \\ a_k=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-jk\omega_0 n} & \text{基频 } \omega_0=\frac{2\pi}{N} \end{cases}$$

CTFT
连续→连续　$x(t)\to X(j\omega)$

$$\begin{cases} X(j\omega)=\int_{-\infty}^{+\infty} x(t)e^{-j\omega t}dt \\ x(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t}d\omega \end{cases}$$

DTFT
离散→连续　$x[n]\to X(e^{j\omega})$

$$\begin{cases} X(e^{j\omega})=\sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n} \\ x[n]=\frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega \end{cases}$$

	CTFS $x(t)$	a_k	DTFS $x[n]$	a_k	CTFT $x(t)$	$X(j\omega)$	DTFT $x[n]$	$X(e^{j\omega})$
线性	$Ax(t)+By(t)$	Aa_k+Bb_k	$Ax[n]+By[n]$	Aa_k+Bb_k	$ax(t)+by(t)$	$aX(j\omega)+bY(j\omega)$	$ax[n]+by[n]$	$aX(e^{j\omega})+bY(e^{j\omega})$
时移	$x(t-t_0)$	$a_k e^{-jk\omega_0 t_0}$	$x[n-n_0]$	$a_k e^{-jk(2\pi/N)n_0}$	$x(t-t_0)$	$X(j\omega)e^{-j\omega t_0}$	$x[n-n_0]$	$X(e^{j\omega})e^{-j\omega n_0}$
频移	$e^{jM\omega_0 t}x(t)$	a_{k-M}	$e^{jM(2\pi/N)n}x[n]$	a_{k-M}	$x(t)e^{j\omega_0 t}$	$X(j(\omega-\omega_0))$	$x[n]e^{j\omega_0 n}$	$X(e^{j(\omega-\omega_0)})$
共轭	$x^*(t)$	a^*_{-k}	$x^*[n]$	a^*_{-k}	$x^*(t)$	$X^*(-j\omega)$	$x^*[n]$	$X^*(e^{-j\omega})$
时反	$x(-t)$	a_{-k}	$x[-n]$	a_{-k}	$x(-t)$	$X(-j\omega)$	$x[-n]$	$X(e^{-j\omega})$
时伸缩	$x(\alpha t)$ $(\alpha>0\ \frac{T}{\alpha})$	a_k	$x_{(m)}[n]=\begin{cases}x[n/m] & n\text{为}m\text{倍数}\\ 0\end{cases}$	$\frac{1}{m}a_k$ (周期为mN)	$x(at)$	$\frac{1}{\lvert a\rvert}X(j\frac{\omega}{a})$	$x_{(k)}[n]=\begin{cases}x[\frac{n}{k}] & n\text{为}k\text{的整数倍}\\ 0\end{cases}$	$X(e^{jk\omega})$
周期卷积	$\int_T x(\tau)y(t-\tau)d\tau$	Ta_kb_k	$\sum_{r=\langle N\rangle} x[r]y[n-r]$	Na_kb_k	$x(t)*y(t)$	$X(j\omega)Y(j\omega)$	$x[n]*y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
时域相乘	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$	$x[n]y[n]$	$\sum_{l=\langle N\rangle} a_l b_{k-l}$	$x(t)y(t)$	$\frac{1}{2\pi}X(j\omega)*Y(j\omega)$	$x[n]y[n]$	$\frac{1}{2\pi}X(e^{j\omega})*Y(e^{j\omega})=\frac{1}{2\pi}\int_{2\pi}X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$
时域微分	$\frac{dx(t)}{dt}$	$jk\omega_0 a_k$	$x[n]-x[n-1]$	$(1-e^{-jk(2\pi/N)})a_k$	$\frac{dx(t)}{dt}$	$j\omega X(j\omega)$	$x[n]-x[n-1]$	$(1-e^{-j\omega})X(e^{j\omega})$
时域积分	$\int_{-\infty}^t x(t)dt$ 只有$a_0=0$时 有限且周期	$(\frac{1}{jk\omega_0})a_k$	$\sum_{k=-\infty}^{n} x[k]$ 只有$a_0=0$时 有限且周期	$(\frac{1}{1-e^{-jk(2\pi/N)}})a_k$	$\int_{-\infty}^t x(\tau)d\tau$	$\frac{X(j\omega)}{j\omega}+\pi X(j0)\delta(\omega)$	$\sum_{k=-\infty}^{n} x[k]$	$\frac{1}{1-e^{-j\omega}}X(e^{j\omega})$
帕斯瓦尔	$\frac{1}{T}\int_T \lvert x(t)\rvert^2dt=\sum_{k=-\infty}^{+\infty}\lvert a_k\rvert^2$		$\frac{1}{N}\sum_{n=\langle N\rangle}\lvert x[n]\rvert^2=\sum_{k=\langle N\rangle}\lvert a_k\rvert^2$		$tx(t)$	$j\frac{dX(j\omega)}{d\omega}$	$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$

$$\int_{-\infty}^{+\infty}\lvert x(t)\rvert^2dt=\frac{1}{2\pi}\int_{-\infty}^{+\infty}\lvert X(j\omega)\rvert^2d\omega \qquad \sum_{n=-\infty}^{+\infty}\lvert x[n]\rvert^2=\frac{1}{2\pi}\int_{2\pi}\lvert X(e^{j\omega})\rvert^2d\omega$$

共轭对称　$x(t)$ real　$a_k=a^*_{-k}$　$\begin{cases}Re\{a_k\}=Re\{a_{-k}\}\\ Im\{a_k\}=-Im\{a_{-k}\}\\ \lvert a_k\rvert=\lvert a_{-k}\rvert\\ \angle a_k=-\angle a_{-k}\end{cases}$

$x(t) \longleftrightarrow X(j\omega)$

实信号 $x(t)=x^*(t)$　共轭对称 $X^*(j\omega)=X(-j\omega)$

共轭对称 $x(t)=x^*(-t)$　实信号 $X^*(j\omega)=X(j\omega)$

对偶：若 $x(t)\xrightarrow{F}X(j\omega)$ 则 $X(t)\xrightarrow{F}2\pi x(-\omega)$

实信号　$x(t)$实偶 $\leftrightarrow a_k$实偶　$x(t)$实奇 $\leftrightarrow a_k$虚奇　$x_e(t)=\frac{x(t)+x(-t)}{2}\leftrightarrow Re\{a_k\}$　$x_o(t)=\frac{x(t)-x(-t)}{2}\leftrightarrow jIm\{a_k\}$

$x(t)$实 $\begin{cases}\text{幅度谱偶 } \lvert X(j\omega)\rvert=\sqrt{Re\{X(j\omega)\}^2+Im\{X(j\omega)\}^2}\\ \text{相位谱奇 } \theta(\omega)=\arctan\left[\frac{Im\{X(j\omega)\}}{Re\{X(j\omega)\}}\right]\end{cases}$

$H(j\omega)=\int_{-\infty}^{+\infty}h(t)e^{-j\omega t}dt$　$j=e^{j\frac{\pi}{2}}$　$-1=e^{j\pi}$

对LTI系统 $\begin{cases}\text{连续 } b_k=a_kH(jk\omega_0)\\ \text{离散 } b_k=a_kH(e^{jk\frac{2\pi}{N}})\end{cases}$

抽样函数 $Sa(x)=\frac{\sin x}{x}$　$sinc(x)=\frac{\sin(\pi x)}{\pi x}$

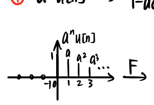
$sgn(x)=\begin{cases}1 & x>0\\ 0 & x=0\\ -1 & x<0\end{cases}$

		CTFT	CTFS
周期	$\sum_{k=-\infty}^{+\infty}a_ke^{jk\omega_0t}$	$2\pi\sum_{k=-\infty}^{+\infty}a_k\delta(\omega-k\omega_0)$	a_k
	$e^{j\omega_0t}$	$2\pi\delta(\omega-\omega_0)$	$a_1=1$
	$\cos(\omega_0t)$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$	$a_1=a_{-1}=\frac{1}{2}$
	$\sin(\omega_0t)$	$\frac{\pi}{j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$	$a_1=-a_{-1}=\frac{1}{2j}$
	$x(t)=1$	$2\pi\delta(\omega)$	$a_0=1$ (T任意)
	$x(t)=\begin{cases}1 & \lvert t\rvert<T_1\\ 0 & T_1<\lvert t\rvert\le\frac{T}{2}\end{cases}$	$\sum_{k=-\infty}^{+\infty}\frac{2\sin k\omega_0T_1}{k}\delta(\omega-k\omega_0)$	$\frac{\omega_0T_1}{\pi}Sa(k\omega_0T_1)=\frac{\sin(k\omega_0T_1)}{k\pi}$
	$p(t)=\sum_{n=-\infty}^{+\infty}\delta(t-nT)$	$\frac{2\pi}{T}\sum_{k=-\infty}^{+\infty}\delta(\omega-\frac{2\pi k}{T})$	$a_k=\frac{1}{T}$
	$x(t)=\frac{E}{T}t$ $(-\frac{T}{2}<t<\frac{T}{2})$		$a_k=\frac{j}{2k\pi}(-1)^kE$
	$x(t)=E(1-\frac{2\lvert t\rvert}{T})$ $(-\frac{T}{2}<t<\frac{T}{2})$		$a_k=\frac{E}{2}Sa^2(\frac{k\pi}{2})$
	$x(t)=\begin{cases}E\cos(\omega_0t) & -\frac{T}{4}\le t\le\frac{T}{4}\\ 0 & \frac{T}{4}<\lvert t\rvert\le\frac{T}{2}\end{cases}$		$a_k=\frac{E}{(1-k^2)\pi}\cos(\frac{k\pi}{2})$
非周期	$x(t)=\begin{cases}1 & \lvert t\rvert<T_1\\ 0 & \lvert t\rvert>T_1\end{cases}$	$\frac{2\sin\omega T_1}{\omega}$	
	$\frac{\sin Wt}{\pi t}$	$X(j\omega)=\begin{cases}1 & \lvert\omega\rvert<W\\ 0 & \lvert\omega\rvert>W\end{cases}$	
	$\delta(t)$	1	
	$\delta(t-t_0)$	$e^{-j\omega t_0}$	
	$u(t)$	$\frac{1}{j\omega}+\pi\delta(\omega)$	
	$e^{-at}u(t)$ $(Re\{a\}>0)$	$\frac{1}{a+j\omega}$	
	$te^{-at}u(t)$ $(Re\{a\}>0)$	$\frac{1}{(a+j\omega)^2}$	
	$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t)$ $(Re\{a\}>0)$	$\frac{1}{(a+j\omega)^n}$	

		DTFT	DTFS
周期	$\sum_{k=\langle N\rangle}a_ke^{jk\omega_0n}$	$2\pi\sum_{k=-\infty}^{+\infty}a_k\delta(\omega-k\omega_0)$	Fourier Series Coefficients (if periodic): a_k
	$e^{j\omega_0n}$	$2\pi\sum_{k=-\infty}^{+\infty}\delta(\omega-\omega_0-2\pi k)$	(a) $\omega_0=\frac{2\pi m}{N}$, $a_k=\begin{cases}1, & k=m,m\pm N,m\pm2N,\ldots\\ 0, & \text{otherwise}\end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational ⇒ The signal is aperiodic
	$\cos(\omega_0n)$	$\pi\sum_{k=-\infty}^{+\infty}\{\delta(\omega-\omega_0-2\pi k)+\delta(\omega+\omega_0-2\pi k)\}$	(a) $\omega_0=\frac{2\pi m}{N}$, $a_k=\begin{cases}\frac{1}{2}, & k=\pm m,\pm m\pm N,\pm m\pm2N,\ldots\\ 0, & \text{otherwise}\end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational ⇒ The signal is aperiodic
	$\sin(\omega_0n)$	$\frac{\pi}{j}\sum_{k=-\infty}^{+\infty}\{\delta(\omega-\omega_0-2\pi k)-\delta(\omega+\omega_0-2\pi k)\}$	(a) $\omega_0=\frac{2\pi r}{N}$, $a_k=\begin{cases}\frac{1}{2j}, & k=r,r\pm N,r\pm2N,\ldots\\ -\frac{1}{2j}, & k=-r,-r\pm N,-r\pm2N,\ldots\\ 0, & \text{otherwise}\end{cases}$ (b) $\frac{\omega_0}{2\pi}$ irrational ⇒ The signal is aperiodic
	$x[n]=1$	$2\pi\sum_{k=-\infty}^{+\infty}\delta(\omega-2\pi k)$	$a_k=\begin{cases}1, & k=0,\pm N,\pm2N,\ldots\\ 0, & \text{otherwise}\end{cases}$
	$x[n]=\begin{cases}1 & \lvert n\rvert\le N_1\\ 0 & N_1<\lvert n\rvert\le N/2\end{cases}$	$2\pi\sum_{k=-\infty}^{+\infty}a_k\delta(\omega-\frac{2\pi k}{N})$	$a_k=\frac{\sin[(2\pi k/N)(N_1+\frac{1}{2})]}{N\sin[2\pi k/2N]}$, $k\ne0,\pm N,\pm2N,\ldots$; $a_k=\frac{2N_1+1}{N}$, $k=0,\pm N,\pm2N,\ldots$
	$\sum_{k=-\infty}^{+\infty}\delta[n-kN]$	$\frac{2\pi}{N}\sum_{k=-\infty}^{+\infty}\delta(\omega-\frac{2\pi k}{N})$	$a_k=\frac{1}{N}$ for all k
非周期	$x[n]=\begin{cases}1 & \lvert n\rvert\le N_1\\ 0 & \lvert n\rvert>N_1\end{cases}$	$\frac{\sin[\omega(N_1+\frac{1}{2})]}{\sin(\omega/2)}$	
	$(0<W<\pi)$ $\frac{\sin Wn}{\pi n}=\frac{W}{\pi}sinc(\frac{Wn}{\pi})$	$X(\omega)=\begin{cases}1 & 0\le\lvert\omega\rvert\le W\\ 0 & W<\lvert\omega\rvert\le\pi\end{cases}$ (周期2π)	
	$\delta[n]$	1	
	$\delta[n-n_0]$	$e^{-j\omega n_0}$	
	$u[n]$	$\frac{1}{1-e^{-j\omega}}+\sum_{k=-\infty}^{+\infty}\pi\delta(\omega-2\pi k)$	
	$a^nu[n]$ $\lvert a\rvert<1$	$\frac{1}{1-ae^{-j\omega}}$	
	$(n+1)a^nu[n]$ $\lvert a\rvert<1$	$\frac{1}{(1-ae^{-j\omega})^2}$	
	$\frac{(n+r-1)!}{n!(r-1)!}a^nu[n]$ $\lvert a\rvert<1$	$\frac{1}{(1-ae^{-j\omega})^r}$	

典型信号DTFT

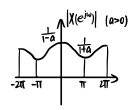
① $a^nu[n]\xrightarrow{F}\frac{1}{1-ae^{-j\omega}}$　$\lvert a\rvert<1$



$X(e^{j\omega})=\frac{1}{1-a\cos\omega+ja\sin\omega}$

$\lvert X(e^{j\omega})\rvert=\frac{1}{\sqrt{1+a^2-2a\cos\omega}}$

$\theta(\omega)=-\arctan\left(\frac{a\sin\omega}{1-a\cos\omega}\right)$

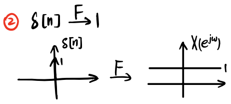


$x(t)=a_0+\sum_{n=1}^{\infty}a_n\cos(n\omega_0t)+\sum_{n=1}^{\infty}b_n\sin(n\omega_0t)$

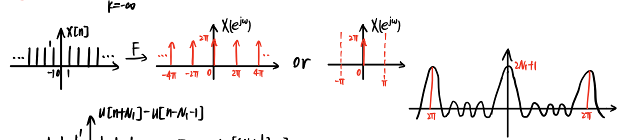
偶 $=a_0+\sum_{n=1}^{\infty}a_n\cos(n\omega_0t)$

奇 $=\sum_{n=1}^{\infty}b_n\sin(n\omega_0t)$

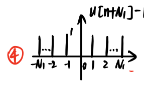
② $\delta[n]\xrightarrow{F}1$



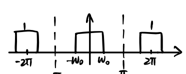
③ $1\xrightarrow{F}2\pi\sum_{k=-\infty}^{+\infty}\delta(\omega-2k\pi)$



④ $u[n+N_1]-u[n-N_1-1]\xrightarrow{F}\frac{\sin[(N_1+\frac{1}{2})\omega]}{\sin(\frac{1}{2}\omega)}=1+2\cos\omega+2\cos2\omega+\ldots+2\cos N_1\omega$



⑤ $\frac{\sin(\omega_0n)}{\pi n}\xrightarrow{F}$ $(0<\omega_0<\pi)$



$=\frac{1}{2}\left[\sum_{n=-\infty}^{+\infty}\frac{\sin(\omega_0+\omega)n}{\pi n}+\sum_{n=-\infty}^{+\infty}\frac{\sin(\omega_0-\omega)n}{\pi n}\right]$

$\begin{cases}\text{当 } -\omega_0<\omega<\omega_0 & =1\\ \text{当 } \omega_0<\omega<\pi & =0\\ \text{当 } -\pi<\omega<-\omega_0 & =0\end{cases}$

当$n=0$时 $x[0]=\lim_{n\to0}\frac{\sin(\omega_0n)}{\pi n}=\frac{\omega_0}{\pi}$

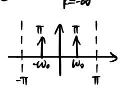
$\sum_{n=-\infty}^{+\infty}\frac{\sin(\omega n)}{\pi n}=\begin{cases}1 & 0<\omega<\pi\\ 0 & \omega=0\\ -1 & -\pi<\omega<0\end{cases}$

$\int_{-\infty}^{+\infty}\frac{\sin(\omega t)}{\pi t}dt=\begin{cases}1 & 0<\omega<\pi\\ 0 & \omega=0\\ -1 & -\pi<\omega<0\end{cases}$

$\frac{\sin(\omega t)}{\pi t}$ 抽样函数

⑥ $u[n]\xrightarrow{F}\frac{1}{1-e^{-j\omega}}+\pi\sum_{k=-\infty}^{+\infty}\delta(\omega-2k\pi)$

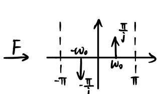
⑦ $\cos(\omega_0n)\xrightarrow{F}$



$=\frac{1}{2}[e^{j\omega_0n}+e^{-j\omega_0n}]$

$\pi\left[\sum_{k=-\infty}^{+\infty}\delta(\omega+\omega_0-2k\pi)+\delta(\omega-\omega_0-2k\pi)\right]$

$\sin(\omega_0n)\xrightarrow{F}$



$\frac{\pi}{j}\left[\sum_{k=-\infty}^{+\infty}\delta(\omega-\omega_0-2k\pi)-\delta(\omega+\omega_0-2k\pi)\right]$

$x(t)=\frac{a_0}{2}+\sum_{n=1}^{\infty}[a_n\cos(n\omega_0t)+b_n\sin(n\omega_0t)]$

$a_0=\frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}}x(t)dt$　　偶 $x(t)=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos(n\omega_0t)$

$a_n=\frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}}x(t)\cos(n\omega_0t)dt$　　奇 $x(t)=\sum_{n=1}^{\infty}b_n\sin(n\omega_0t)$

$b_n=\frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}}x(t)\sin(n\omega_0t)dt$

Tip $\frac{1}{j\omega+a}\leftrightarrow e^{-at}u(t)$

$\frac{1}{(j\omega+a)^n}\leftrightarrow\frac{t^{n-1}}{(n-1)!}e^{-at}u(t)$

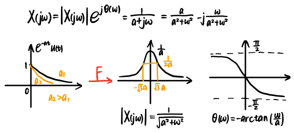
$x(t)*\delta(t-t_0)=x(t-t_0)$

$x(t)\cdot\delta(t-t_0)=x(t_0)\cdot\delta(t-t_0)$

典型信号 CTFT

① $e^{-at}u(t)\xrightarrow{F}\frac{1}{a+j\omega}$ $(a>0)$ 只对$a>0$成立

$X(j\omega)=\lvert X(j\omega)\rvert e^{j\theta(\omega)}=\frac{1}{a+j\omega}=\frac{a}{a^2+\omega^2}-j\frac{\omega}{a^2+\omega^2}$

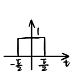


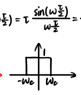
$\lvert X(j\omega)\rvert=\frac{1}{\sqrt{a^2+\omega^2}}$　　$\theta(\omega)=-\arctan(\frac{\omega}{a})$

② $\delta(t)\xrightarrow{F}1$　③ $1\xrightarrow{F}2\pi\delta(\omega)$　⑥ $u(t)\xrightarrow{F}\frac{1}{j\omega}+\pi\delta(\omega)$

$\lim_{\omega\to+\infty}\frac{\sin(\omega t)}{t}=\pi\delta(t)$

$x(t)$ 乘π / 除π $X(j\omega)$

④  $\xrightarrow{F}\tau Sa(\frac{\omega\tau}{2})=\tau\frac{\sin(\frac{\omega\tau}{2})}{\frac{\omega\tau}{2}}=\frac{2\sin(\frac{\omega\tau}{2})}{\omega}$

⑤ $\frac{\sin(\omega_ct)}{\pi t}=\frac{\omega_c}{\pi}Sa(\omega_ct)\xrightarrow{F}$ 

方波与Sa 互为傅利叶变换

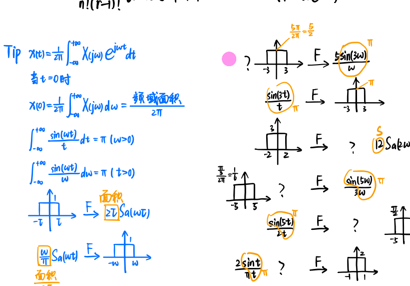
Tip $x(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}X(j\omega)e^{j\omega t}dt$

当$t=0$时

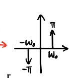
$x(0)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}X(j\omega)d\omega=\frac{\text{频谱面积}}{2\pi}$

$\int_{-\infty}^{+\infty}\frac{\sin(\omega t)}{t}dt=\pi$ $(\omega>0)$

$\int_{-\infty}^{+\infty}\frac{\sin(\omega t)}{\omega}d\omega=\pi$ $(t>0)$



⑦ $\cos(\omega_0t)\xrightarrow{F}$  $\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$

$\sin(\omega_0t)\xrightarrow{F}$  $\frac{\pi}{j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$

离散 LTI系统的卷积

①列表法

已知 $x[n]=3\delta[n+1]+2\delta[n]+\delta[n-1]-\delta[n-2]$

$h[n]=\delta[n+1]+\delta[n]+\delta[n-1]-\delta[n-2]$

求 $y[n]=x[n]*h[n]$

$x\backslash h$	1	1	2	-1
3	3	3	6	-3
2	2	2	4	-2
1	1	1	2	-1
-1	-1	-1	-2	1
	3 5 9	1	-3	1

②卷积公式法

$h[n]$ 1 1 2 -1

$h[-n]$ -1 2 1 1

$x[n]$ 3 2 1 -1

-1 2 1 1

3=3

-1 2 1 1

3+2=5

-1 2 1 1

6+2+1=9

-1 2 1 1

-3+4+1-1=1

注意：①是h[n]翻转 不是x[n]翻转

②翻转 准平移准

$x[n]*h[n]=\sum_{k=-\infty}^{+\infty}x[k]\cdot h[-(k-n)]$

相乘 求和 翻转 平移

给一堆条件求 x(t)

(1) ① $a_k=a_{k+2}$ → 令 $m=k-2$ $\quad x(t)=\sum_{m=-\infty}^{+\infty}a_{m+2}e^{j(m+2)\frac{2\pi}{T}t}=\sum_{m=-\infty}^{+\infty}a_m e^{jm\frac{2\pi}{T}t}$

② $a_k=a_{-k}$ → $x(t)$偶 $\quad \Rightarrow x(t)=x(t)e^{j\frac{4\pi}{T}t}$ $\quad x(t)=0$ 或 $e^{j\frac{4\pi}{T}t}=1 \Rightarrow t=\frac{T}{2}n$

设 $x(t)=A\delta(t)+B\delta(t-\frac{3}{2})$ $\quad 0<t<3$

③ $\int_{-0.5}^{0.5}x(t)dt=1 \rightarrow \int_{-0.5}^{0.5}A\delta(t)dt=1 \rightarrow A=1$

④ $\int_1^2 x(t)dt=2 \rightarrow \int_1^2 B\delta(t-\frac{3}{2})dt=2 \rightarrow B=2$

$x(t)=\sum_{k=-\infty}^{+\infty}\delta(t-3k)+2\delta(t-3k-1.5)$

$T=3 \quad \omega_0=\frac{2\pi}{3} \quad x(t)=\sum_{k=-\infty}^{+\infty}a_k e^{jk\frac{2\pi}{3}t}$

只在有限孤立点有值的函数 积分值为0

(2) ① $x(t)$ is real → $a_{-k}=a_k^*$

② $a_k=0$ for $k=0$ and $k>2$ → $a_1\ a_{-1}\ a_2\ a_{-2}$

③ $x(t)=-x(t-3)$ → $a_k e^{-jk\omega_0 3}=a_k(-1)^k \quad a_k=-[a_k(-1)^k]$

k偶 $a_k=-a_k$ $\quad a_{\pm2}=0$ $\quad$ k奇 $a_1=-a_{-1}=a$

$x(t)=ae^{j\frac{\pi}{3}t}-ae^{-j\frac{\pi}{3}t}=2a\cos(\frac{\pi}{3}t)$

④ $\frac{1}{6}\int_{-3}^{3}|x(t)|^2dt=\frac{1}{2} \rightarrow \sum_{k=-\infty}^{+\infty}|a_k|^2=\frac{1}{2} \quad a=\frac{1}{2}$

⑤ $T=6 \quad \omega_0=\frac{\pi}{3}$ $\quad \therefore x(t)=\cos(\frac{\pi}{3}t)$

⑥ a_1 为正实数

(3) ① $Re\{X(j\omega)\}=0$ → $x(t)$ real and odd

② $Im\{X(j\omega)\}=0$ → $x(t)$ real and even $\quad e^{j\omega t_0}X(j\omega)\leftrightarrow x(t+t_0)$

③ 存在实数α 使 $e^{j\alpha\omega}X(j\omega)$为实数 $\quad Im\{e^{j\alpha\omega}X(j\omega)\}=0$ $\quad t=-\alpha$ 处有对称轴

④ $\int_{-\infty}^{\infty}X(j\omega)d\omega=0$ → $x(0)=0$

⑤ $\int_{-\infty}^{\infty}\omega X(j\omega)d\omega=0$ → $x'(t)\leftrightarrow j\omega X(j\omega)$ $\quad x'(0)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}j\omega X(j\omega)d\omega$ $\quad x'(0)=0$

⑥ $X(j\omega)$ is periodic → $x(t)$ is discrete

只满足①③⑤

对偶性

(1) $e^{-|t|}\leftrightarrow\frac{2}{1+\omega^2}$ $\quad f(t)\leftrightarrow F(\omega)$

(1) 求 $te^{-|t|}\leftrightarrow j\frac{d}{d\omega}(\frac{2}{1+\omega^2})=-\frac{4j\omega}{(1+\omega^2)^2}$ $\quad F(t)\leftrightarrow 2\pi f(-\omega)$

(2) 求 $\frac{4t}{(1+t^2)^2}$ $\quad$ 令 $F(t)=-\frac{4jt}{(1+t^2)^2}\leftrightarrow 2\pi f(\omega)=-2\pi\omega e^{-|\omega|}$

$\frac{4jt}{(1+t^2)^2}\leftrightarrow 2\pi\omega e^{-|\omega|}$

$\left[\frac{\sin(\frac{\pi}{4}n)}{\pi n}\right]^2 \xrightarrow{F} ?$

$\frac{1}{2\pi}$ $\quad \frac{1}{4}$

$8\left(\frac{\sin(\frac{\pi}{8}n)}{\pi n}\right)^2 \xrightarrow{F} ?$

$\frac{1}{2\pi}$ $\quad A=8 \quad B=1$

若AB=1 令B=1 A=$\frac{1}{8}$

$8(-1)^n\left(\frac{\sin(\frac{\pi}{8}n)}{\pi n}\right)^2 \xrightarrow{F} ?$

$(-1)^n$可以把低通滤波器变成高通滤波器

$\frac{\sin(\omega_c n)}{\pi n}\xrightarrow{F}$ 低通

$(-1)^n\frac{\sin(\omega_c n)}{\pi n}\xrightarrow{F}$ 高通（变化更剧烈）

连续 LTI系统的卷积

$x(t)=e^{-bt}u(t) \quad h(t)=e^{-at}u(t)$ 求 $x(t)*h(t)$

直接代入公式 $x(t)*h(t)=\int_{-\infty}^{+\infty}x(\tau)h(t-\tau)d\tau$

$x(t)*h(t)=\int_{-\infty}^{+\infty}x(\tau)h(t-\tau)dt$

$=\int_{-\infty}^{+\infty}e^{-b\tau}u(\tau)e^{-a(t-\tau)}u(t-\tau)dt$

仅当 $0<\tau<t$时 $u(\tau)u(t-\tau)=1$ 其他都为0

若$t<0$时 $x(t)*h(t)=0$

而当$t>0$时 $x(t)*h(t)=\int_0^t e^{-b\tau}e^{-a(t-\tau)}d\tau=\frac{e^{-at}-e^{-bt}}{b-a}$ $\quad (t>0)$

求

当$0<t<1$时 $y(t)=\frac{(2-t)t}{2}$ $\quad$ 当$1<t<2$时 $y(t)=\frac{(2-t)^2}{2}$ $\quad$ 当$t>2$ 或 $t<0$时 $y(t)=0$

检验看临界点

除非 x(t)或 h(t)有δ函数，否则 x(t)*h(t)不会跳变

两个宽度一样的方波的卷积是三角波

两个宽度不一样的方波的卷积是梯形波

$|x|x|=1 \quad 2x|x|+|x|2x|=0 \quad 2x|2|x|=-4 \quad 0$

当$0<t<1$时 面积 $[\frac{1}{2}t^2+\frac{1}{2}(t-2)^2]-[1-\frac{1}{2}t^2-\frac{1}{2}(t-2)^2]=t^2+(t-2)^2-1$

当$1<t<2$时 面积 $[1-\frac{1}{2}(t-1)^2-\frac{1}{2}(t-2)^2]-[\frac{1}{2}(t-1)^2+\frac{1}{2}(t-2)^2]=1-(t-1)^2-(t-2)^2$

综上 $y(t)=\begin{cases}t^2+(t-2)^2-1 & 0<t<1\\ 1-(t-1)^2-(t-2)^2 & 1<t<2\end{cases}$

翻转

求 $x(\frac{3-t}{2})=x(-\frac{1}{2}(t-3))$

$\delta(at+b)=\frac{1}{|a|}\delta(t+\frac{b}{a})$ 拉伸

若 $x(t)=2\delta(t-1)$

则 $x(\frac{3-t}{2})$

$=2\delta(\frac{3-t}{2}-1)$

$=2\delta(-\frac{1}{2}t+\frac{1}{2})$

$=2\frac{1}{|-\frac{1}{2}|}\delta(t-1)=4\delta(t-1)$

右移

$2\pi\left(\frac{\sin(\frac{3}{4}\pi n)}{\pi n}\right)^2 \xrightarrow{F}$

法2° 简便算法：①算直接卷积

②结果以2π为周期重复

③重合处相加

法3° $\left(\frac{\sin(\frac{3}{4}\pi n)}{\pi n}\right)^2=\left(\frac{\sin(\frac{1}{4}\pi n)}{\pi n}\right)^2+\frac{1}{2}\delta[n]$

四个不常用信号的傅利叶变换

① 双边指数信号 $x(t)=e^{-a|t|} \xrightarrow{F} \frac{2a}{a^2+\omega^2}$

② 高斯脉冲信号 $x(t)=Ee^{-(\frac{t}{\tau})^2} \xrightarrow{F} \sqrt{\pi}E\tau e^{-(\frac{\omega\tau}{2})^2}$

③ 半波余弦信号 $x(t)=\begin{cases}E\cos(\frac{\pi t}{\tau}) & |t|<\frac{\tau}{2}\\ 0 & |t|>\frac{\tau}{2}\end{cases} \xrightarrow{F} \frac{2E\tau}{\pi}\frac{\cos(\frac{\omega\tau}{2})}{1-(\frac{\omega\tau}{\pi})^2}$

④ 奇对称斜线 $x(t)=\frac{E}{T}t \ (-T<t<T) \xrightarrow{F} j\frac{E}{T}\left[\frac{2\sin(\omega T)}{\omega^2}-\frac{2T}{\omega}\cos(\omega T)\right]$

频域的线性相位 对应时域延时

求 $X(j\omega)$反变换 $x(t)$ $\quad |X(j\omega)|e^{j2\omega}$

$\frac{\sin[2(t-2)]}{\pi(t-2)} \xrightarrow{F}$

三角面积 $\xrightarrow{F} 2Sa^2(\omega)$

$Sa(\omega)\cdot Sa(\omega)$

$\xrightarrow{F} \frac{E\tau}{2}Sa^2(\frac{\omega\tau}{4})$

$X(j0)=\int_{-\infty}^{+\infty}x(t)dt$

$X(j\omega)=ASa^2(\omega)$

$X(j0)=ASa^2(0)=A=\int_{-\infty}^{+\infty}x(t)dt$

用傅利叶变换解电路图

输入 $V(t)$ $\quad R \quad C \quad V_c(t)$ 输出

$u_R(t)=Ri_R(t)$

$U_R(t)=RI_R(t)$

$u_c(t)=\frac{1}{C}\int_{-\infty}^{t}i_c(\tau)d\tau$

$=\frac{1}{C}i(t)*u(t)$

$U_c(t)=\frac{1}{C}I_c(j\omega)[\frac{1}{j\omega}+\pi\delta(\omega)]$

$=\frac{1}{j\omega C}I_c(j\omega)+\frac{\pi}{C}I_c(j0)\delta(\omega)$ $\quad$ 0

$u_L(t)=L\frac{du(t)}{dt}$

$U_L(t)=j\omega L\ I_L(j\omega)$

已知 $\frac{1}{RC}=2$ 输入 $x(t)=u(t)$

$H(j\omega)=\frac{V_c(j\omega)}{V(j\omega)}=\frac{\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}}=\frac{1}{j\omega RC+1}=\frac{2}{j\omega+2}$

$V_c(j\omega)=H(j\omega)X(j\omega)$

$=\frac{2}{j\omega+2}[\frac{1}{j\omega}+\pi\delta(\omega)]=(\frac{1}{j\omega}+\pi\delta(\omega))-\frac{1}{j\omega+2}$

$v_c(t)=u(t)-e^{-2t}u(t)$

求 CTFT

(1) $x(t)=e^{-|t|}\cos(2t)$

$X(j\omega)=\frac{1}{2\pi}\left[\frac{2}{1+\omega^2}\right]*\left[\pi(\delta(\omega-2)+\delta(\omega+2))\right]$

$=\frac{1}{1+(\omega-2)^2}+\frac{1}{1+(\omega+2)^2}$

(2) $x(t)=\frac{\sin(2\pi t)}{\pi(t-1)}=\frac{\sin(2\pi(t-1))}{\pi(t-1)}$

$y(t)=\frac{\sin(2\pi t)}{\pi t}\leftrightarrow Y(j\omega)=u(\omega+2\pi)-u(\omega-2\pi)$

$x(t)=y(t-1)\leftrightarrow X(j\omega)=e^{-j\omega}[u(\omega+2\pi)-u(\omega-2\pi)]$

(3) $x(t)=\begin{cases}t^2 & 0<t<1\\ 0 & 其它\end{cases}$

$y(t)=u(t)-u(t-1)$ $\quad u(t)\leftrightarrow\frac{1}{j\omega}+\pi\delta(\omega)$

$Y(j\omega)=(1-e^{-j\omega})(\frac{1}{j\omega}+\pi\delta(\omega))$ $\quad t^n x(t)\leftrightarrow j^n\frac{d^n}{d\omega^n}X(j\omega)$

$X(j\omega)=j^2\frac{d^2}{d\omega^2}Y(j\omega)$

(4) $x_4(t)=(1-|t|)u(t+1)u(1-t)=\begin{cases}1+t & -1<t\le 0\\ 1-t & 0<t<1\\ 0 & 其它\end{cases}$

$x_4'(t)=\begin{cases}1 & -1<t\le0\\ -1 & 0<t<1\\ 0 & 其它\end{cases}=u(t+1)-2u(t)+u(t-1)$

$x_4''(t)=\delta(t+1)-2\delta(t)+\delta(t-1)$ $\quad \frac{d^n x(t)}{dt^n}\leftrightarrow(j\omega)^n X(j\omega)$

$F\{x_4''(t)\}=e^{j\omega}-2+e^{-j\omega}=2\cos\omega-2=(j\omega)^2X(j\omega)$

$a_k=\frac{1}{N}\sum_{n=0}^{N-1}x[n]e^{-jk(2\pi/N)n}$ $\quad a_k=\frac{1}{N}\sum_{m=0}^{M-1}\left(\sum_{n=mN/M}^{(m+1)N/M-1}x[n]e^{-jk\frac{2\pi}{N}n}\right)$

$n'=n-m\frac{N}{M}$

$a_k=\frac{1}{N}\sum_{m=0}^{M-1}\sum_{n'=0}^{\frac{N}{M}-1}x[n'+m\frac{N}{M}]e^{-jk(\frac{2\pi}{N})n'}e^{-jk(\frac{2\pi}{N})m\frac{N}{M}}$

$=\frac{1}{N}\sum_{m=0}^{M-1}\sum_{n'=0}^{\frac{N}{M}-1}x[n'+m\frac{N}{M}]e^{-jk(\frac{2\pi}{N})n'}e^{-j\frac{2\pi km}{M}}$

$=\frac{1}{N}\sum_{n'=0}^{\frac{N}{M}-1}e^{-jk\frac{2\pi}{N}n'}\left[\sum_{m=0}^{M-1}x[n'+m\frac{N}{M}]e^{-j\frac{2\pi km}{M}}\right]$

$k=pM$ $\quad \sum_{m=0}^{M-1}x[n'+m\frac{N}{M}]=0$

Problem 8

(c) $X(e^{j\omega})$ is purely imaginary

If $X(e^{j\omega})$ is purely imaginary, then its corresponding time-domain signal $x[n]$ must be a real and odd signal $x[n]=-x[-n]$

(d) $e^{j\omega}X(e^{j\omega})$ is purely real for some integer k

If $X(e^{j\omega})$ is purely real, then its corresponding time-domain signal $x[n]$ must be a real and even signal $x[n]=x[-n]$

$e^{jk\omega}X(e^{j\omega})\leftrightarrow x[n+k]$ must be a real and even

$\frac{(t+1)X_1}{2}Sa(2\omega)Sa(\omega)$

法1° $Sa^2(\frac{\omega}{2})[1+e^{j\omega}+e^{-j\omega}+\dots]$

$\xrightarrow{F} Sa^2(\frac{\omega}{2})[e^{-j3\omega}+e^{j3\omega}]$

$=2Sa^2(\frac{\omega}{2})\cos(3\omega)$

法2° $=Sa(\frac{\omega}{2})e^{j\frac{3}{2}\omega}[\frac{1}{j\omega}+\pi\delta(\omega)]$

$\frac{\sin(t-1)}{\pi(t-1)} \xrightarrow{F}$

$\frac{\sin(t-1)}{\pi(t-1)}2\cos(3t)$

$e^{-a|t|}\xrightarrow{F}\frac{2a}{a^2+\omega^2}$

① $\frac{1}{1+jt}\xrightarrow{F}?$ $\quad$ ③ $\frac{1}{\pi t}\xrightarrow{F}?$

$e^{-t}u(t)\xrightarrow{F}\frac{1}{1+j\omega}$ $\quad$ 先求 ? $\xrightarrow{F}\frac{1}{\pi\omega}$

$\frac{1}{1+jt}\xrightarrow{F}2\pi e^{\omega}u(-\omega)$ $\quad u(t)\xrightarrow{F}\frac{1}{j\omega}+\pi\delta(\omega)$

② $\frac{2}{t^2+1}\xrightarrow{F}?$ $\quad \frac{1}{2}\xrightarrow{F}\pi\delta(\omega)$ $\quad$ 90°移相器

$e^{-|t|}\xrightarrow{F}\frac{2}{\omega^2+1}$ $\quad \frac{1}{j}[u(t)-\frac{1}{2}]\xrightarrow{F}\frac{1}{\pi\omega}$

$\frac{2}{t^2+1}\xrightarrow{F}2\pi e^{-|\omega|}$ $\quad \frac{1}{\pi t}\xrightarrow{F}j[u(-\omega)-\frac{1}{2}]$

周期变换

? $\xrightarrow{F}$ $\quad$ 以π为周期

$\frac{\sin(\frac{\pi}{4}n)+\sin(\frac{3}{4}\pi n)}{\pi n}\xrightarrow{F}$ $\quad$ 以2π为周期

$\frac{\sin(\frac{3}{4}\pi n)}{\pi n}\xrightarrow{F}$

$\frac{\sin(\frac{\pi}{4}n)}{\pi n}\xrightarrow{F}$

$X(e^{j\omega})$

$\xrightarrow{F^{-1}}x[n]$

n偶 变为2x[n] +

$[1+(-1)^n]\frac{\sin(\frac{3}{4}\pi n)+\sin(\frac{\pi}{4}n)}{\pi n}$ $\quad$ n奇 变为0 $\quad X(e^{j(\omega-\pi)})$

$x[n]$ $\quad \xrightarrow{F^{-1}}(-1)^n x[n]$

? $\xrightarrow{F}$

高通滤波器

法1°

$x[n]=\delta[n]-\frac{\sin(\frac{\pi}{4}n)}{\pi n}$

法2° $x[n]=\frac{\sin(\frac{3}{4}\pi n)}{\pi n}\xrightarrow{F}$ $\quad X(e^{j\omega})$

$(-1)^n\frac{\sin(\frac{\pi}{4}n)}{\pi n}\xrightarrow{F}$ $\quad X(e^{j(\omega-\pi)})$

三角函数

$\cos(\frac{5}{4}\pi n)$ $\xrightarrow{F}$

$=\cos(-\frac{3}{4}\pi n)=\cos(\frac{3}{4}\pi n)$

$\sin(\frac{5}{4}\pi n)$ $\xrightarrow{F}$

$=\sin(-\frac{3}{4}\pi n)=-\sin(\frac{3}{4}\pi n)$

$\cos(\pi n)$ $\xrightarrow{F}$

$\sin(\pi n)\xrightarrow{F}0$

$\cos(\pi n)=e^{j\pi n}=e^{-j\pi n}=(-1)^n$

$\cos(\pi n)=e^{j\pi n}\ X(e^{j\omega})=2\pi\sum_{k=-\infty}^{+\infty}\delta(\omega-(2k+1)\pi)$

CTFS

1. $a_k = \begin{cases} \frac{(-1)^{k-1}}{k^2\pi^2} & k\neq 0 \\ \frac{1}{2} & k=0 \end{cases}$ $\begin{cases} 0 & k偶 \\ \frac{-2}{k^2\pi^2} & k奇 \end{cases}$

$x(t) \xrightarrow{h(t)} y(t) = 1 - \cos(\frac{3\pi}{2}t)$

求 $A_1 A_2 A_3$ 及 $\Omega_1 \Omega_2 \Omega_3$

$y(t) = \sum_{k=-\infty}^{+\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$ $\omega_0 = \frac{2\pi}{T} = \frac{\pi}{2}$

直流分量 $(k=0)$ $a_0 H(j0) = 1 \Rightarrow H(j0) = 2$ $A_1 = 2$

基波分量 $(k=\pm1)$ $H(j\frac{\pi}{2}) = 0$ $Y(j\omega) = X(j\omega)H(j\omega)$

三次谐波分量 $(k=\pm3)$ $a_3 H(j\frac{3\pi}{2}) = -\frac{1}{2} \Rightarrow H(j\frac{3\pi}{2}) = \frac{9\pi^2}{4}$ $A_1 = A_3 = \frac{9\pi^2}{4}$

$0 < \Omega_1 < \frac{\pi}{2}$ $\frac{\pi}{2} < \Omega_2 < \frac{3\pi}{2} < \Omega_3 < \frac{5\pi}{2}$

DTFT

2. (1) $X(e^{j0}) = \sum_{n=-\infty}^{\infty} x[n] = -1+1+2+1\ldots = 6$

(2) $X(e^{j\pi}) = \sum_{n=-\infty}^{\infty} x[n](-1)^n$

$= (-1)(-1)^3 + 1(-1)^2 + 2(-1)^1 + 1(-1)^0 + 0 + 1(-1)^2 + 2(-1)^3 \ldots$

(1) $\omega = 0$ 时 $X(n) = \sum_{n=-\infty}^{+\infty} x[n]$

$X_{(0)} = (-1) + 2 + 0 + 2 + (-1) = 2$

(2) $x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n} d\omega$

$\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega = 2\pi X[0] = 2\pi \cdot 0 = 0$

(3) $\sum_{n=-\infty}^{+\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{2\pi} |X(e^{j\omega})|^2 d\omega$ $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega = 2\pi \sum |x[n]|^2 = 2\pi(1+1+2^2+1^2)$

CTFT

3. (1) 求 $\int_{-\infty}^{+\infty} \text{sinc}^4(t) dt$ $\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$

$\text{sinc}(t) \cdot \text{sinc}(t) =$

$\int_{-\infty}^{+\infty} \text{sinc}^4(t) dt = \frac{1}{2\pi}\int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega = \frac{1}{2\pi}\int_{-2\pi}^{2\pi} (1-\frac{|\omega|}{2\pi})^2 d\omega = \frac{2}{3}$

(2) $= u(t+1) - 1 + u(t-1)$

求 FT

$= (\frac{1}{j\omega} + \pi\delta(\omega))e^{j\omega} - 2\pi\delta(\omega) + (\frac{1}{j\omega} + \pi\delta(\omega))e^{-j\omega}$

$= \frac{2}{j\omega}\cos(\omega) + \pi\delta(\omega)(e^{j\omega} + e^{-j\omega} - 2)$

$= \frac{2}{j\omega}\cos(\omega) + \pi\delta(\omega)(e^{j0} + e^{-j0} - 2) = \frac{2}{j\omega}\cos(\omega)$

CTFT

4. ① x is real

② $x(t) = 0$ $t \leq 0$ $t>0$ $x(-t) = 0$

③ $F^{-1}\{Re\, X(j\omega)\} = |t|e^{-|t|}$ $= \frac{x(t)+x(-t)}{2}$ $\frac{x(t)}{2} = te^{-t}$ $\Rightarrow x(t) = 2te^{-t}u(t)$

采样

5. zero-hold

$T = 秒 1\,ms$

(1) ZOH 处理：输入冲激序列 $x(t) = \sum x[n]\delta(t-nT)$

输出阶梯波 $y(t) = \sum x[n]p(t-nT)$

线性：$x_1(t) = \sum x_1[n]\delta(t-nT) + \sum x_2[n]\delta(t-nT)$ $x_4(t) = a x_1(t)$

$\Rightarrow y_3(t) = \sum(x_1[n] + x_2[n])p(t-nT)$ $\Rightarrow Y_4(t) = a y(t)$

$= y_1(t) + y_2(t)$

时不变：$X_{new}(t) = x(t-kT) = \sum x[n]\delta(t-(n+k)T) = \sum_m x[m-k]\delta(t-mT)$

$Y_{new}(t) = \sum_m x[m-k]p(t-mT) = \sum x[n]p(t-(n+k)T) = y(t-kT)$

对整数倍 T 时不变，任意微小时变

(2) 求 system 冲激响应

$\delta(t) \to$ 保持 T $h(t) = u(t) - u(t-T) = \begin{cases} 1 & 0<t<T \\ 0 & 其它 \end{cases}$

(3) 求 system function

$H(j\omega) = (\frac{1}{j\omega} + \pi\delta(\omega)) - (\frac{1}{j\omega} + \pi\delta(\omega))e^{-j\omega T}$

$= \frac{1}{j\omega}(1 - e^{-j\omega T}) = T\frac{\sin(\omega T/2)}{\omega T/2}e^{-j\omega T/2}$

调制

6. $x(t)$ $H(j\omega)$ $\cos(\omega_c t)$ $\sin(\omega_c t)$ $y(t)$ $H(j\omega) = \begin{cases} -j & \omega>0 \\ j & \omega<0 \end{cases}$ $h(t)$ 实

(1) $x(t)$ real 则 $y(t)$ real

real 充要条件 (频域) 有共轭对称性 $H^*(j\omega) = H(-j\omega)$

(2) 从 $y(t)$ 中重建 $x(t)$ $y(t) \frac{\sin(\omega_c t)}{}$ 低通滤波 $\to \times 2$

$y(t) = [x(t) * h(t)]\cos(\omega_c t) + x(t)\sin(\omega_c t)$

$y(t)\sin\omega_c t = [x(t)*h(t)]\cos\omega_c t \sin\omega_c t + x(t)\sin^2\omega_c t$

$= \frac{1}{2}[x(t)*h(t)]\sin 2\omega_c t + \frac{1}{2}x(t) - \frac{1}{2}x(t)\cos 2\omega_c t$

基带

2. $H(e^{j\omega}) = 1 + 2\cos(\omega/2)$ 求 $\int_{-\pi}^{\pi} X(e^{j\omega})H(e^{j\omega})e^{j6\omega} d\omega$

DTFT $\quad 2\pi y[6] \leftarrow Y(e^{j\omega})$

$h[n] = \delta[n] + \delta[n-1] + \delta[n+1]$ $y[6] = x[5] + x[6] + x[7]$

$y[n] = x[n] * h[n] = x[n] + x[n-1] + x[n+1]$

3. $x(t)$ is periodic in $T = 1s$ sampling with 6 samples per sec

采样 $x[n]$ is periodic in $N = 6$ $x[n] = \sum_{r=0}^{5} c_r e^{jr\frac{2\pi}{6}n}$

① $x[n]$ symmetric of n $\to x[5] = x[1]$ $x[2] = x[4]$

② $x[n]$ positive of all n $\to x[0] + 2x[1] + 2x[2] + x[3] = 3$ ①

③ $x[0] + x[1] + \ldots + x[5] = 3$ $\to x[0] - 2x[1] + 2x[2] - x[3] = 1$ ②

④ $x[0] - x[1] + x[2] - x[3] + x[4] - x[5] = 1$

⑤ most FS 系数为 0 至少有2个不为0

①+② $\Rightarrow x[0] + 2x[2] = 2$ $\Rightarrow x[n] = C_0 + C_3(-1)^n$ $\begin{cases} x[0] = x[2] = C_0 + C_3 \\ x[1] = x[3] = C_0 - C_3 \end{cases}$

①-② $\Rightarrow 2x[1] + x[3] = 1$

$\Rightarrow C_0 = 0.5$ $C_3 = \frac{1}{6}$ $x[n] = 0.5 + \frac{1}{6}(-1)^n$

构造 $x(t)$ 使采样点 $t = \frac{n}{6}$ 处 满足 $x(\frac{n}{6}) = 0.5 + \frac{1}{6}\cos(\pi n)$

① 无混叠最小带宽

$\Omega = \pi = \omega T_s = 6\pi$ $f_s = 6Hz$ $x_1(t) = 0.5 + \frac{1}{6}\cos(6\pi t)$

② 用高频混叠 $\cos$ 频率增加 6Hz 整数倍

$x_2(t) = 0.5\cos(12\pi t) + \frac{1}{6}\cos(6\pi t)$

$\cos(2\pi n) = 1$ 常数换余弦

1. 求 $x(t) = 2\cos(\frac{\pi}{2}t) + 4\cos(\frac{\pi}{3}t)$ 的 FS 系数和基波周期 T

CTFS $T_1 = 4s$ $T_2 = 6s$ $T = LCM(4,6) = 12$ $\omega_0 = \frac{2\pi}{T} = \frac{\pi}{6}$ rad/s

$x(t) = 2\cos(3\omega_0 t) + 4\cos(2\omega_0 t)$

$= e^{j3\omega_0 t} + e^{-j3\omega_0 t} + 2e^{j2\omega_0 t} + 2e^{-j2\omega_0 t}$

$\frac{e^{j\omega} - e^{-j\omega}}{j\omega} = e^{-j\frac{1}{2}\omega}\frac{e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}}{j\omega} = e^{-j\frac{1}{2}\omega}\frac{\sin(\omega/2)}{\omega/2}$

4. LTI CT system pole-zero diagram:

CTFS s-plane

Assume $x(t) = \begin{cases} 3 & |t| \leq 1 \\ 0 & 1<|t|<4 \end{cases}$

fundamental $T = 8$ input: a_k output: b_k

If $\frac{b_0}{a_0} = 1$ determine $\frac{b_1}{a_1}$

① $H(s) = K\frac{s-(-1)}{(s-j)(s+j)} = K\frac{s+1}{s^2+1}$

② $\omega_0 = \frac{2\pi}{T} = \frac{\pi}{4}$

对 LTI 系统 $b_k = a_k H(jk\omega_0) \Rightarrow \frac{b_k}{a_k} = H(jk\omega_0)$

$\frac{b_0}{a_0} = H(0) = K = 1$ $H(s) = \frac{s+1}{s^2+1}$

③ $\frac{b_1}{a_1} = H(j\frac{\pi}{4}) = \frac{\frac{j\pi}{4}+1}{(\frac{j\pi}{4})^2+1}$

3. causal and stable signal $x(t)$ has FT

CTFT $X(j\omega) = \frac{j\omega}{(j\omega+1)(2j\omega+1)}$

(1) 求 $X(j\omega)$ 的 inverse FT

(2) 求 $\frac{1}{2\pi}\int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega$

(1) ~~$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t}dt$~~ 典型信号

$X(j\omega) = \frac{1}{j\omega+1} + \frac{-1}{2j\omega+1} = \frac{1}{j\omega+1} + \frac{-\frac{1}{2}}{j\omega+\frac{1}{2}}$

$x(t) = e^{-t}u(t) + (-\frac{1}{2})e^{-\frac{1}{2}t}u(t)$

(2) $\frac{1}{2\pi}\int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega = \int_{-\infty}^{+\infty} |x(t)|^2 dt$

$= \int_0^{+\infty} |e^{-t} - \frac{1}{2}e^{-\frac{1}{2}t}|^2 dt$

$= \int_0^{+\infty} (e^{-2t} - e^{-\frac{3}{2}t} + \frac{1}{4}e^{-t}) dt = \frac{1}{12}$

$\int_0^{+\infty} e^{-at} dt = -\frac{1}{a}e^{-at}\Big|_0^{+\infty} = -\frac{1}{a}(0-1) = \frac{1}{a}$

6. 调制 $\omega_c \gg \omega_m$

$\cos(\frac{1}{2}\omega_m t)$ $\cos(\omega_c t + \frac{1}{2}\omega_m t)$ W_1 W_2 LPF V_1 V_2 X Y $\sin(\frac{1}{2}\omega_m t)$ $\sin(\omega_c t + \frac{1}{2}\omega_m t)$

Assume LPF is ideal cutoff $\omega_m/2$

Input has FT as following

$X(j\omega)$ 求 $Y(j\omega)$

$\cos\omega_c t$ $\sin\omega_c t$ $-\omega_c$ ω_c

1. DTFS DT LTI System $x[n] = \cos(\frac{\pi n}{2})$

$y[n] - \frac{1}{4}y[n-1] = x[n] + 2x[n-4]$

求 output 的 Fourier series representation

① 频域求 DTFT $H(e^{j\omega}) = \frac{1+2e^{-j4\omega}}{1-\frac{1}{4}e^{-j\omega}}$

② $x[n]$ 基波周期 $N = 4$ 基频 $\omega_0 = \frac{\pi}{2}$

$x[n] = \frac{1}{2}e^{j\frac{\pi}{2}n} + \frac{1}{2}e^{-j\frac{\pi}{2}n}$

$a_0 = 0$ $a_1 = \frac{1}{2}$ $a_2 = 0$ $a_3 = \frac{1}{2}$

③ $b_k = a_k H(e^{jk\omega_0})$

$H(e^{jk\frac{\pi}{2}}) = \frac{1+2e^{-jk2\pi}}{1-\frac{1}{4}e^{-jk\frac{\pi}{2}}}$

5. 采样 $X_p(j\omega)$ $x_p(t)$ Impulse to Sequence $x[n]$ $X(e^{j\Omega})$

$x(t)$ $X(j\omega)$ $p(t) = \sum_{k=-\infty}^{+\infty} \delta(t-kT)$ $T = 0.5 \times 10^{-3}\, s$

(1) $X_p(j\omega) = \frac{1}{T}\sum_{k=-\infty}^{+\infty} X(j(\omega - k\omega_s))$

① $\omega_s = \frac{2\pi}{T} = 4\pi \times 10^3$ rad/s $= 2\omega_m$ $\frac{1}{T} = 2000$

② 将 $X_p(j\omega)$ 频率轴 ω 归一化到 $\Omega = \omega T$

$\Omega = 0, \pm 2\pi$, 正负...

(2) ① $X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t}dt$

$X(j0) = \int_{-\infty}^{+\infty} x(t)dt$ 由 $X(j\omega)$ $X(j0) = 1$

② $X(e^{j\Omega}) = \sum_{-\infty}^{+\infty} x[n]e^{-j\Omega n}$

$X(e^{j0}) = \sum_{-\infty}^{+\infty} x[n]$ 由 $X(e^{j\Omega})$ $X(e^{j0}) = 2000$

$x(t) = \begin{cases} 1 & |t|<T_1 \\ 0 & T_1<|t|<T/2 \end{cases}$ $\leftrightarrow a_k = \frac{\sin(k\omega_0 T_1)}{k\pi}$ $a_0 = 1$

(1) $a_k = \begin{cases} 0 & k=0 \\ (j)^k \frac{\sin k\pi/4}{k\pi} & 其它 \end{cases}$ $T = 4$ $\omega_0 = \frac{2\pi}{T} = \frac{\pi}{2}$

Sa(x)

$\frac{k\pi}{4} = k\omega_0 T_1 \Rightarrow T_1 = \frac{1}{2}$

$y(t) = \begin{cases} 1 & |t|<\frac{1}{2} \\ 0 & \frac{1}{2}<|t|<2 \end{cases}$ $\leftrightarrow b_k = \frac{\sin k\pi/4}{k\pi}$

$b_0 = \frac{1}{4}$ $a_0 = 0$ $\Rightarrow p(t) = y(t) - \frac{1}{4}$ 分段

$a_k = b_k(e^{j\frac{\pi}{2}})^k$ $j\frac{\pi}{2}k = -jk\omega_0 t_0$ $\Rightarrow t_0 = -1$

$x(t) = p(t - t_0) = p(t+1) = \begin{cases} \frac{3}{4} & -\frac{3}{2}<t<-\frac{1}{2} \\ -\frac{1}{4} & -3<t<-\frac{3}{2},\ -\frac{1}{2}<t<1 \end{cases}$

(4) 带数 $a_k = \begin{cases} 1 & k偶 \\ 2 & k奇 \end{cases}$ $x(t) = \delta(t-kT) \to a_k = \frac{1}{T}$

$T = 4$ $\omega_0 = \frac{2\pi}{T} = \frac{\pi}{2}$

$a_k = b_k + c_k$ $x(t) = C \cdot T\sum_{n=-\infty}^{+\infty}\delta(t-nT) \leftrightarrow a_k = C$

$b_k = 1$ for all k $\leftrightarrow y(t) = \sum_{k=-\infty}^{+\infty} 4\delta(t-4k)$

$C_k = \begin{cases} 0 & k偶 \\ 1 & k奇 \end{cases}$ $\leftrightarrow z(t) = \sum_{k=奇} 1 \cdot e^{jk\frac{\pi}{2}t} = \sum_{m=-\infty}^{+\infty} e^{j(2m+1)\frac{\pi}{2}t}$

$\sum_{k=-\infty}^{+\infty}\delta(t-kT) = \frac{1}{T}\sum_{k=-\infty}^{+\infty} e^{jk\omega_0 t}$ $= e^{j\frac{\pi}{2}t}\sum_{m=-\infty}^{+\infty} e^{jm\pi t} = 2\sum_{m=-\infty}^{+\infty} e^{j\frac{\pi}{2}t}\delta(t-2m)$

$= 2(-1)^k\delta(t-2k)$

$\Rightarrow x(t) = y(t) + z(t) = \sum_{k=-\infty}^{+\infty} 4\delta(t-4k) + 2(-1)^k\delta(t-2k)$

$x(t)$ $\cos\omega_c t$ $s(t)$ $p(t)$ $y(t)$ $H(j\omega)$ $v(t)$ 方波调制

$\cos\omega_c t$ $p(t)$ 1/ω_c

Re Im ω_m

① (1) $s(t)$ Re Im $\frac{1}{2}$

② $p(t)$ $p(t) = \begin{cases} +1 & \frac{\pi}{2}<\theta<\frac{\pi}{2} \\ -1 & \frac{\pi}{2}<\theta<\frac{3\pi}{2} \end{cases}$ $P(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos n\theta$ $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} P(\theta)\cos n\theta\, d\theta = \frac{4}{n\pi}\sin\frac{n\pi}{2}$

$p(t) = \frac{4}{\pi}[\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \frac{1}{5}\cos 5\omega_c t \ldots]$

$P(j\omega) = 4\sum_{m=0}^{\infty}\frac{(-1)^m}{2m+1}[\delta(\omega-(2m+1)\omega_c) + \delta(\omega+(2m+1)\omega_c)]$ $P(j\omega)$ 4 $-\frac{4}{3}$

③ $y(t) = x(t)\cos\omega_c t \cdot \text{sgn}(\cos\omega_c t) = x(t)|\cos\omega_c t|$

$|\cos\omega_c t| = \frac{2}{\pi}X(j\omega) + \frac{2}{3\pi}[X(j(\omega - 2\omega_c)) + X(j(\omega + 2\omega_c))] - \frac{2}{15\pi}[X(j(\omega - 4\omega_c)) + X(j(\omega + 4\omega_c))] + \ldots$

Re $\frac{2}{\pi}$ Im $\frac{2}{3\pi}$

(2) 基带分量 $\frac{2}{\pi}X(j\omega)$ $V(j\omega) = H(j\omega)Y(j\omega) = X(j\omega) \Rightarrow v(t) = x(t)$

$H(j\omega) = \begin{cases} \frac{\pi}{2} & |\omega|<\omega_c \\ 0 & |\omega|>\omega_c \end{cases}$

频分复用

$y(t) \to H_1(j\omega) \to \otimes \to H_2(j\omega) \to r(t)$ $\cos(\omega_c t + \omega_p)t$ $z(t)$

(1) $y_1(t) = x_1(t)\cos\omega_c t$ $Y_1(j\omega) = \frac{1}{2}X_1(j(\omega-\omega_c)) + \frac{1}{2}X_1(j(\omega+\omega_c))$

$\to H_1$ 后 $KY_1(j\omega)$ $\cos\omega_c t\cos(\omega_c + \omega_p)t = \frac{1}{2}\cos\omega_p t + \frac{1}{2}\cos(2\omega_c + \omega_p)t$

$\times\cos(\omega_c + \omega_p)t \to z(t) = \frac{K}{2}x_1(t)\cos\omega_p t + \frac{K}{2}x_1(t)\cos(2\omega_c + \omega_p)t$

$A(j\omega)$ ω_c $\omega_c + \omega_m$ $\omega_c + \omega_T$ $Z(j\omega)$ ω_p $\omega_p + 2\omega_c$

(2) $\begin{cases} 2\omega_c + \omega_p - \omega_T > \omega_p + \omega_m \\ \omega_T > \omega_m \end{cases}$ $H(j\omega)$

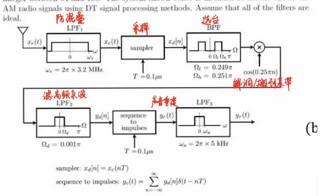
(3) $H_2(j\omega)$ 保留中心 $[\omega_p - \omega_m, \omega_p + \omega_m]$ α β

过 H_1 混频后 目标项为 $\frac{K}{2}x_1(t)\cos\omega_p t$

$r(t) = x_1(t)\cos\omega_p t$ $G \cdot \frac{K}{2} = 1$

Problem 1

Commercial AM radio stations broadcast radio frequencies within a limited range: $2\pi(f_c - 5\text{ kHz}) < \omega < 2\pi(f_c + 5\text{ kHz})$, where $f_c = \omega_c/(2\pi) = n \times 10$ kHz and n is an integer between 54 and 160. The system shown below is intended to decode one of the AM radio signals using DT signal processing methods. Assume that all of the filters are ideal.



sampler: $x_d[n] = x_c(nT)$

sequence to impulses: $y_c(t) = \sum_{n=-\infty}^{\infty} y_d[n]\delta(t - nT)$

Part a. Determine the center frequency f_c for the AM station that this receiver will detect.

Part b. Which of the following statement(s) is/are correct?

b1. Increasing the cutoff frequency ω_r of LPF$_1$ by a factor of 1.5 will cause aliasing.

b2. Decreasing the cutoff frequency ω_r of LPF$_1$ by a factor of 2 will have no effect on the output $y_r(t)$.

b3. Halving the sampling interval T would have no effect on the output $y_r(t)$.

b4. Doubling the sampling interval T would have no effect on the output $y_r(t)$.

Part c. Which of the following statement(s) is/are correct?

c1. Increasing the cutoff frequency Ω_d of LPF$_2$ will change $y_r(t)$ by adding signals from unwanted radio stations.

c2. Increasing the cutoff frequency Ω_d of LPF$_2$ will change $y_r(t)$ because aliasing will occur.

c3. Doubling the cutoff frequency Ω_d of LPF$_2$ will have no effect on $y_r(t)$.

c4. Halving the cutoff frequency Ω_d of LPF$_2$ will have no effect on $y_r(t)$.

(a)

$\Omega = \omega T = 2\pi f T$

带通滤波器的中心频率为: $\Omega_c = \frac{\Omega_l + \Omega_h}{2} = \frac{0.249\pi + 0.251\pi}{2} = 0.25\pi$

采样周期 $T = 0.1\mu s = 10^{-7}$s $\quad 0.25\pi = 2\pi f_c \times 10^{-7}$

$f_c = \frac{0.25}{2 \times 10^{-7}} = 1.25 \times 10^6\text{Hz} = 1.25\text{MHz}$ $\quad \Omega = \omega T = 2\pi f T$ T 为采样周期

$f_c = n \times 10\text{kHz}$ $\quad n = 125$

(b) $f_s = \frac{1}{T} = 10\text{MHz}$ $\quad f_N = \frac{f_s}{2} = 5\text{MHz}$

LPF1 的截止频率 ω_r 增大为原来的 1.5 倍，会导致混叠 **b1 错误** 截止频率为 $f_r = 3.2$MHz

截止频率提高 1.5 倍后，新截止频率为 $3.2 \times 1.5 = 4.8$MHz

$4.8\text{MHz} < f_N = 5\text{MHz}$ 在奈奎斯特频率以内，因此不会引起混叠

LPF1 的截止频率 ω_r 减小为原来的一半，不会影响输出 $y_r(t)$ **b2 正确** 截止频率减半后，新截止频率为 $3.2 \times 0.5 = 1.6$MHz

当前目标检测的中心频率是 1.25MHz，其高频边界只有 1.255MHz。低于 1.6MHz，对最终输出 $y_r(t)$ 没有影响。

将采样间隔 T 减半翻倍，不会影响输出 $y_r(t)$。 **b3 b4 错误**

采样间隔减半后，$T' = 0.05\mu s$ $\quad \Omega' = 2\pi f_c T' = 2\pi(1.25 \times 10^6)(0.5 \times 10^{-7}) = 0.125\pi$

采样间隔翻倍后，$T' = 0.2\mu s$ $\quad \Omega' = 2\pi f_c T' = 2\pi(1.25 \times 10^6)(2 \times 10^{-7}) = 0.5\pi$

带通滤波器仍然固定在：$0.249\pi \sim 0.251\pi$

采样间隔 T 决定了 ω 到 Ω 的映射尺度，因此会对输出 $y_r(t)$ 产生影响，导致无法正确解调当前信号。

(c) 信号通过 BPF 后，再乘以 $\cos(0.25\pi n)$ 进行数字相乘解调，频谱会发生左右平移，搬移 $\pm 0.25\pi$。原本位于 $\pm 0.25\pi$ 的目标信号被搬移回中心基带 $\Omega = 0$ 附近，带宽为 $\pm 0.001\pi$。同时，还会产生一组位于 $\pm 0.5\pi$ 处的高频倍频分量。

LPF$_2$ 的截止频率是 0.001π，保留基带信号。

增大 LPF$_2$ 的截止频率 Ω_d，会把不需要的广播电台信号加入进来，从而改变 $y_r(t)$。 **c1 错误**

增大 LPF$_2$ 的截止频率不会加入其他不想要的信号。因为其他信号已经在前面的 BPF 中被滤掉。LPF$_2$ 后面即使放宽，也不会加入其他信号。

增大 LPF$_2$ 的截止频率 Ω_d，会因为发生混叠而改变 $y_r(t)$。 **c2 错误**

混叠发生在采样阶段，由采样间隔 T 和模拟输入频率决定。增大 LPF$_2$ 的截止频率，只是数字域内的低通滤波变换，不涉及降低采样率的操作，因此不会发生混叠。

将 LPF$_2$ 的截止频率 Ω_d 加倍，不会影响 $y_r(t)$。 **c3 正确**

Ω_d 翻倍变成 0.002π，目标基带信号（带宽 0.001π）能通过，搬移产生的倍频干扰（在 $\pm 0.5\pi$ 附近）远远高于 0.002π，仍会被 LPF2 阻挡。LPF$_2$ 的截止频率加倍，对 $y_r(t)$ 没有影响。

将 LPF$_2$ 的截止频率 Ω_d 减半，不会影响 $y_r(t)$。 **c4 错误**

把 LPF$_2$ 的截止频率减半，对 $y_r(t)$ 有影响。将 Ω_d 减半，变成 0.0005π，它会去除目标基带信号的一半带宽，导致输出的音频信号失真。

采样

采样频率(采样率) $f_s=8Hz$ (1秒采样8个)
采样角频率 $\omega_s=2\pi f_s$
采样周期 $T_s=\frac{1}{f_s}$
Nyquist Rate: $\omega_s=2\omega_m$
Nyquist Frequency: $f_N=\frac{f_s}{2}$ (最高ω无混叠)

连续时间信号的采样

1. 冲激串采样(理想采样)

$x_p(t)=x(t)p(t)=\sum_{n=-\infty}^{\infty}x(nT)\delta(t-nT)$

$X_p(j\omega)=\frac{1}{T}\sum_{k=-\infty}^{\infty}X(j(\omega-k\omega_s))$ 采样角频率 $\omega_s=\frac{2\pi}{T}$

2. 零阶保持器(ZOH)采样

$h_0(t)=u(t)-u(t-T)$

$H_0(j\omega)=Te^{-j\omega T/2}\frac{\sin(\omega T/2)}{\omega T/2}=Te^{-j\omega T/2}Sa(\frac{\omega T}{2})$

$X_0(j\omega)=X_p(j\omega)H_0(j\omega)=e^{-j\omega T/2}Sa(\frac{\omega T}{2})\sum_{k=-\infty}^{\infty}X(j(\omega-k\omega_s))$

3. 采样定理: 连续带限 $x(t)$ 最高截止频率 ω_m

最低采样频率 $\omega_s>2\omega_m$

最大采样周期 $T<\frac{\pi}{\omega_m}$

$x(t)=x_1(t)+x_2(t)$ $\omega_m=\max(\omega_1,\omega_2)$

$x(t)=x_1(t)\cdot x_2(t)$ $\omega_m=\omega_1+\omega_2$

信号重构: 连续 ⇌ 离散

1. 理想重构低通滤波器 $\omega_c=\frac{\omega_s}{2}$

$H_r(j\omega)=\begin{cases}T & |\omega|<\frac{\omega_s}{2}\\ 0 & |\omega|>\frac{\omega_s}{2}\end{cases}$ $\omega_m<\omega_c<\omega_s-\omega_m$

$h_r(t)=\frac{\sin(\pi t/T)}{\pi t/T}=\mathrm{sinc}(\frac{\pi t}{T})$

2. 时域理想重构

$x(t)=x_p(t)*h_r(t)=\sum_{n=-\infty}^{\infty}x(nT)\frac{\sin[\frac{\pi}{T}(t-nT)]}{\frac{\pi}{T}(t-nT)}$

3. C/D 转换

$x[n]=x_c(nT)$ 数字角频率 $\Omega=\omega T=2\pi\frac{\omega}{\omega_s}$ rad/sample rad/s

$X(e^{j\Omega})=\frac{1}{T}\sum_{k=-\infty}^{\infty}X_c(j\frac{\Omega-2\pi k}{T})=\frac{1}{T}X_c(j\omega)\big|_{\omega=\frac{\Omega}{T}}$

标准采样 DTFT 原连续 CTFT

$X_p(j\omega)=X_p(j\frac{\Omega}{T})=X(e^{j\Omega})$ 只是横轴缩放3T

4. D/C 转换

$Y_c(j\omega)=\begin{cases}TY(e^{j\omega T}) & |\omega|<\frac{\pi}{T}\\ 0 & |\omega|>\frac{\pi}{T}\end{cases}$

离散时间信号的采样

1. 抽取 (Decimation) M倍下采样

$y[n]=x[nM]$ 等效采样率变为 $\frac{1}{M}$

$Y(e^{j\Omega})=\frac{1}{M}\sum_{k=0}^{M-1}X(e^{j(\frac{\Omega-2\pi k}{M})})$ 展宽会导致频谱混叠

故抽取前要加 $\omega_c<\frac{\pi}{M}$ 低通滤波器

2. 插值 (Interpolation) L倍上采样

$x_e[n]=\begin{cases}x[n/L] & n=kL\\ 0 & else\end{cases}$ 等效采样率变为 $\frac{1}{L}$

$X_e(e^{j\Omega})=X(e^{j\Omega L})$ 频谱压缩会导致 $[-\pi,\pi]$ 内形成镜像

故插值后要加 $\omega_c=\frac{\pi}{L}$ 低通滤波器

$Y_e(e^{j\Omega})=H(e^{j\Omega})X_e(e^{j\Omega})$

其中 $H(e^{j\Omega})=\begin{cases}L & |\Omega|<\frac{\pi}{L}\\ 0 & \frac{\pi}{L}<|\Omega|\le\pi\end{cases}$

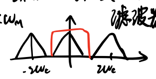
(1) 调制 $y(t)=x(t)\cos(\omega_c t+\theta_c)$

解调 $w(t)=x(t)\cos(\omega_c t+\theta_c)\cos(\omega_c t+\theta_c)$

$=\frac{1}{2}x(t)+\frac{1}{2}x(t)\cos(2\omega_c t+2\theta_c)$

(2) 基带 $|\omega|<\omega_m$ 滤波器输出 $\frac{1}{2}x(t)$ 与 θ_c 无关

高频段



$\omega_{co}>\omega_m$, $\omega_{co}<2\omega_c-\omega_m$

⇒ $\omega_m\le\omega_{co}<2\omega_c-\omega_m$ ⇒ $\omega_c>\omega_m$

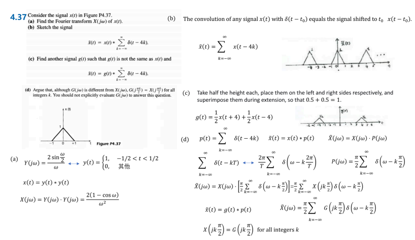
三角公式

$\sin A\cos B=\frac{1}{2}[\sin(A+B)+\sin(A-B)]$

$\cos A\sin B=\frac{1}{2}[\sin(A+B)-\sin(A-B)]$

$\cos A\cos B=\frac{1}{2}[\cos(A+B)+\cos(A-B)]$

$\sin A\sin B=\frac{1}{2}[\cos(A-B)-\cos(A+B)]$



调制

正弦振幅调制(AM)

复指数载波 $y(t)=x(t)e^{j\omega_c t}$

$Y(j\omega)=X(j(\omega-\omega_c))$

正弦载波 $y(t)=x(t)\cos(\omega_c t)$

$Y(j\omega)=\frac{1}{2}X(j(\omega-\omega_c))+\frac{1}{2}X(j(\omega+\omega_c))$

常规调制 $y(t)=[A+x(t)]\cos(\omega_c t)$

$Y(j\omega)=\pi A[\delta(\omega-\omega_c)+\delta(\omega+\omega_c)]+\frac{1}{2}X(j(\omega-\omega_c))+\frac{1}{2}X(j(\omega+\omega_c))$

脉冲串载波 $y(t)=x(t)c(t)$

$c(t)=\sum_{k=-\infty}^{\infty}a_k e^{jk\omega_c t}$ $a_k=\frac{\Delta}{T}Sa(\frac{k\omega_c\Delta}{2})$ $a_0=\frac{\Delta}{T}$

$Y(j\omega)=\sum_{k=-\infty}^{\infty}a_k X(j(\omega-k\omega_c))$

相干/同步解调

① × 载波 $w(t)=y(t)\cos(\omega_c t)=\frac{1}{2}x(t)+\frac{1}{2}x(t)\cos(\omega_c t)$

$W(j\omega)=\frac{1}{2}X(j\omega)+\frac{1}{4}X(j(\omega-2\omega_c))+\frac{1}{4}X(j(\omega+2\omega_c))$

② LPF $H(j\omega)=\begin{cases}2 & |\omega|<\omega_c^{LPF}\\ 0 & |\omega|>\omega_c^{LPF}\end{cases}$

$\omega_m<\omega_c^{LPF}<2\omega_c-\omega_m$

频分复用 FDM

$y(t)=\sum x_i(t)\cos(\omega_i t)$

不混叠条件 ∀两路 i,j $|\omega_i-\omega_j|>\omega_{mi}+\omega_{mj}$

带通滤波器 以 ω_i 为中心, 带宽为 $2\omega_{mi}$

$H_{BPF}(j\omega)=\begin{cases}1 & \omega_i-\omega_{mi}<|\omega|<\omega_i+\omega_{mi}\\ 0 & else\end{cases}$

单边带振幅调制 SSB

(-) 上边带 $Y_{USB}(j\omega)=Y_{DSB}(j\omega)\cdot H_{HPF}(j\omega)$

(+) 下边带 $Y_{LSB}(j\omega)=Y_{DSB}(j\omega)\cdot H_{LPF}(j\omega)$

移相法 $y_{SSB}(t)=x(t)\cos(\omega_c t)\mp\hat{x}(t)\sin(\omega_c t)$

离散时间正弦振幅调制

$y[n]=x[n]\cos(\Omega_c n)$

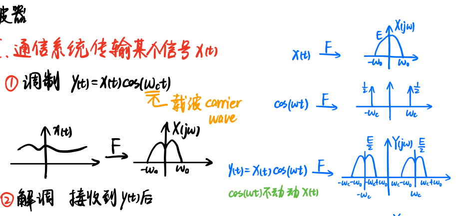
$Y(e^{j\Omega})=\frac{1}{2}X(e^{j(\Omega-\Omega_c)})+\frac{1}{2}X(e^{j(\Omega+\Omega_c)})$

解调不混叠条件 $\Omega_c>\Omega_m$ 且 $\Omega_c+\Omega_m<\pi$

Ⅰ. 通信系统传输某个信号 $x(t)$

① 调制 $y(t)=x(t)\cos(\omega_c t)$

载波 carrier wave



$y(t)=x(t)\cos(\omega_c t)$ $\cos(\omega_c t)$不动 动 $x(t)$

② 解调 接收到 $y(t)$ 后

$W(t)=y(t)\cos(\omega_c t)$

$=x(t)\cos^2(\omega_c t)$

$=\frac{1}{2}x(t)+\frac{1}{2}x(t)\cos(2\omega_c t)$

被低通滤波器滤掉

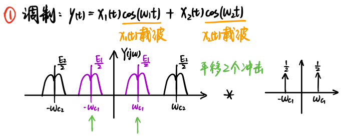
$W(t)=y(t)\cos(\omega_c t)$ $=x(t)\cos^2(\omega_c t)$ $y(t)$不动 动 $\cos(\omega_c t)$

$\omega_m<\omega_{lp}<2\omega_c-\omega_m$

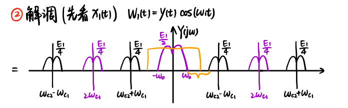
Ⅱ. 通信系统传输两个信号 $x_1(t)$ $x_2(t)$

① 调制: $y(t)=x_1(t)\cos(\omega_1 t)+x_2(t)\cos(\omega_2 t)$

$x_1(t)$载波 $x_2(t)$载波 平移2个冲击

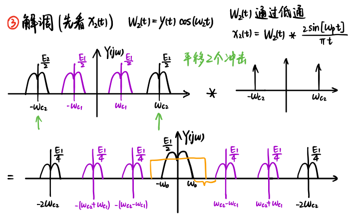


② 解调 (先看 $x_1(t)$) $W_1(t)=y(t)\cos(\omega_1 t)$



③ 解调 (先看 $x_2(t)$) $W_2(t)=y(t)\cos(\omega_2 t)$ $W_2(t)$ 通过低通 $x_2(t)=W_2(t)*\frac{2\sin(\omega_2 t)}{\pi t}$

平移2个冲击



混叠公式

$f_{observed}=|f_{harmonic}-kf_s|$

(1) $x_1(t)=\cos(3000t)$ $x_1[n]=x_1(nT)=\cos(3n)$ $\Omega_1=3$ rad/sample $<\pi$

$x_2(t)=\cos(5000t)$ $x_2[n]=x_2(nT)=\cos(5n)$ $\Omega_2=5$ rad/sample $>\pi$

$T=0.001$

$\cos(\Omega n)$ 主值频率范围 $0\le\Omega\le\pi$ 判断采样后是否混叠

① Ω超出 $x(t)$ 的主值频率区间 (数字,时域)

② $\omega_s<2\omega_m$ ω_m为 $x(t)$最高频率 (连续,频域)

混叠 无 $\Omega_{eff,1}=3$; 有 $\Omega_{eff,2}=2\pi-5\approx1.28$

(2) $x_d[n]=\begin{cases}x_c(nT) & n\ even\\ -x_c(nT) & n\ odd\end{cases}$ $x_d[n]=(-1)^n x_c(nT)=e^{j\pi n}x[n]$

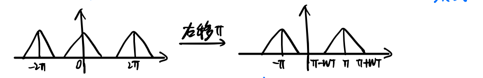
C/D (1)

$X_d(e^{j\Omega})=X(e^{j(\Omega-\pi)})$

$X(e^{j\Omega})=\frac{1}{T}\sum X_c(\frac{\Omega-2\pi k}{T})$

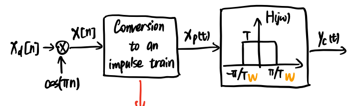
$X_d(e^{j\Omega})=\frac{1}{T}\sum X_c(\frac{\Omega-\pi-2\pi k}{T})$

标准采样: 2π周期延拓, 中心0, ±2π, ±4π… 延拓周期 时域 ω_s 频域 2π



D/C (2) $\pi-WT\ge0$ ⇒ $W\le\frac{\pi}{T}$ 信号可重建 ⇔ 不混叠

(3)



冲激串转换器 $x_p(t)=\sum_{n=-\infty}^{\infty}x[n]\delta(t-nT)=\sum_{n=-\infty}^{\infty}x_c(nT)\delta(t-nT)$

$X_p(j\omega)=\frac{1}{T}\sum_{k=-\infty}^{+\infty}X_c(j(\omega-k\frac{\pi}{T}))$

(3)

$X_c(j\omega)=\begin{cases}1-\frac{|\omega|}{100} & |\omega|\le100\\ 0 & |\omega|>100\end{cases}$

uniform sampler ×$\frac{1}{T}$

① 采样: $X_c(j\omega)\to X_d(e^{j\Omega})$ $x_p(t)=x_c(t)\cdot p(t)=x_c(t)\sum_{n=-\infty}^{+\infty}\delta(t-nT)$

$T=\frac{\pi}{100}$ $\omega_s=\frac{2\pi}{T}=200$ rad/s $=2W$ 恰不混叠

$X_d(e^{j\Omega})=\frac{1}{T}X_c(j\omega)\big|_{\omega=\frac{\Omega}{T}}=\frac{100}{\pi}(1-\frac{|\Omega|}{\pi})$ $|\Omega|\le\pi$

② 数字滤波: $X_d(e^{j\Omega})\to Y_d(e^{j\Omega})$

$X_d(j\omega)=\begin{cases}\frac{100}{\pi}(1-\frac{|\Omega|}{\pi}) & |\Omega|<\frac{\pi}{2}\\ 0 & \frac{\pi}{2}<|\Omega|\le\pi\end{cases}$

③ 冲激重建: $Y_d(e^{j\Omega})\to Y_p(j\omega)$ $y_p(t)=\sum_{n=-\infty}^{\infty}y_d[n]\delta(t-nT)$

sample to impulse

$y_p(t)=\sum_{n=-\infty}^{\infty}y_d[n]\delta(t-nT)$ $Y_p(j\omega)=Y_d(e^{j\omega T})$

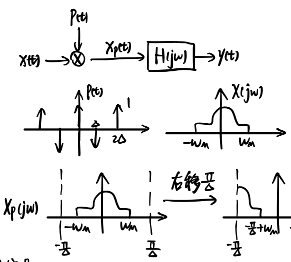
$Y_p(j\omega)=\begin{cases}\frac{100}{\pi}(1-\frac{|\omega|}{100}) & |\omega|\le50\\ 0 & 50<|\omega|\le100\end{cases}$

④ 连续低通滤波: $Y_p(j\omega)\to Y_c(j\omega)$

×T 通带范围 $-100<\omega<100$ 增益 $T=\frac{\pi}{100}$

$Y_c(j\omega)=\begin{cases}1-\frac{|\omega|}{100} & |\omega|\le50\\ 0 & 50<|\omega|\le100\end{cases}$

(4)



(1) 求 $X_p(j\omega)$ 右移 $\frac{\pi}{\Delta}$

$\omega_s=\frac{2\pi}{T}=\frac{2\pi}{2\Delta}$ 采样时刻 $t=n\Delta$ $(-1)^n=e^{j\pi n}=e^{j\frac{\pi}{\Delta}t}$

(1) $P(t)$ 为 $T=2\Delta$ 的周期信号

其 $a_k=\frac{1}{T}\int_T P(t)e^{-jk\frac{2\pi}{T}t}dt=\frac{1}{2\Delta}\int_{-\frac{\Delta}{2}}^{\frac{3\Delta}{2}}P(t)e^{-jk\frac{\pi}{\Delta}t}dt$

$=\frac{1}{2\Delta}\int_{-\frac{\Delta}{2}}^{\frac{3\Delta}{2}}[\delta(t)-\delta(t-\Delta)]e^{-jk\frac{\pi}{\Delta}t}dt$

$=\frac{1}{2\Delta}(1-e^{-jk\pi})=\begin{cases}\frac{1}{\Delta} & k为奇\\ 0 & k为偶\end{cases}$

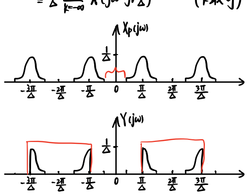
故 $P(j\omega)=\sum_{k=-\infty}^{\infty}2\pi a_k\delta(\omega-k\omega_0)$

$=\sum_{k=-\infty}^{\infty}2\pi a_k\delta(\omega-k\frac{\pi}{\Delta})$

$=\sum_{k=-\infty}^{\infty}2\pi\cdot\frac{1}{\Delta}\cdot\delta(\omega-k\frac{\pi}{\Delta})$ (k取奇)

⇒ $X_p(j\omega)=\frac{1}{2\pi}X(j\omega)*\sum_{k=-\infty}^{\infty}2\pi\frac{1}{\Delta}\cdot\delta(\omega-k\frac{\pi}{\Delta})$ (k取奇)

$=\frac{1}{\Delta}\sum_{k=-\infty}^{\infty}X(j\omega-jk\frac{\pi}{\Delta})$ (k取奇)



(2) $H_0(j\omega)=\begin{cases}\Delta & |\omega|<\omega_m\\ 0 & |\omega|>\omega_m\end{cases}$

$x_p(t)\cdot\cos(\frac{\pi}{\Delta}t)\xleftrightarrow{FT}\frac{1}{2}X_p(j\omega+j\frac{\pi}{\Delta})+\frac{1}{2}X_p(j\omega-j\frac{\pi}{\Delta})$

⇒ $X_r(j\omega)=[\frac{1}{2}X_p(j\omega+j\frac{\pi}{\Delta})+\frac{1}{2}X_p(j\omega-j\frac{\pi}{\Delta})]H_0(j\omega)=X(j\omega)$ ⇒ $x_r(t)=x(t)$

$x_p(t)\to\otimes\to H_0(j\omega)\to x_r(t)$, $\cos(\frac{\pi}{\Delta}t)$

(3) $H_c(j\omega)=\begin{cases}2\Delta & |\omega|<\omega_m\\ 0 & |\omega|>\omega_m\end{cases}$

$y(t)\cos(\frac{\pi}{\Delta}t)\leftrightarrow\frac{1}{2}Y(j\omega+j\frac{\pi}{\Delta})+\frac{1}{2}Y(j\omega-j\frac{\pi}{\Delta})$

⇒ $X_r(j\omega)=[\frac{1}{2}Y(j\omega+j\frac{\pi}{\Delta})+\frac{1}{2}Y(j\omega-j\frac{\pi}{\Delta})]H_c(j\omega)=X(j\omega)$ ⇒ $x_r(t)=x(t)$

$y(t)\to\otimes\to H_c(j\omega)\to x_r(t)$, $\cos(\frac{\pi}{\Delta}t)$

(4) $x_p(t)\to x(t)$ $\frac{\pi}{\Delta}-\omega_m\ge-\frac{\pi}{\Delta}+\omega_m$ 采样

$y(t)\to x(t)$ $\frac{\pi}{\Delta}+\omega_m\le\frac{3\pi}{\Delta}-\omega_m$ ⇒ $\Delta\le\frac{\pi}{\omega_m}$ 滤波